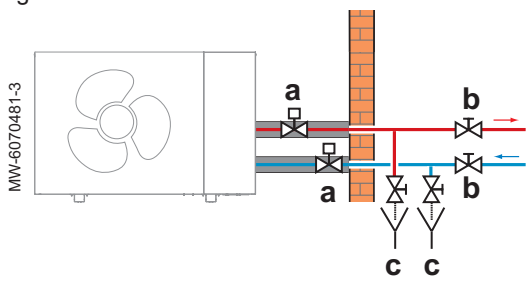
Fig.42



The anti-freeze valves must have the following specifications:

- Opening of the valves at a heating water temperature of +3 °C or less
- Sufficient flow rate to drain the installation before it can freeze

See
Installation manual for anti-freeze valves

The automatic draining solution must be equipped with two isolation valves and two drain valves used to drain the outdoor part of the heating circuit.

a Anti-freeze valve
b Isolation valve
c Drain valve

Caution
In the event of a prolonged power cut (10 hours or more) with negative outdoor temperatures, manual draining is required.

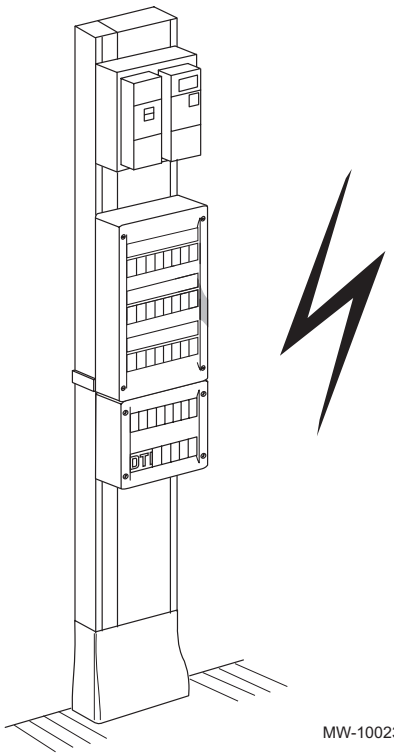
6.9 Electrical connections

Danger
Risk of electrocution.

6.9.1 Checking and preparing the electrical installation

Caution
Only a qualified professional is authorised to work on the electrical part of the installation.

Fig.43



1. Always switch off the electrical installation before making any connections.
2. Respect the requirements of the prevailing standards when selecting the cables and circuit breakers.

Tab.28 Prevailing standards

Country	Standard
Great Britain	BS 7671

3. Check the electrical specifications of the mains power supply available and compare these to the specifications indicated on the data plates on the appliances. The electrical specifications must be compatible.
4. Read and follow the instructions in the manual and the electrical diagrams supplied with the appliance.
5. Select the cables used for the various connections. The cable cross-sections must:
   - Meet the needs of the installation
   - Conform to the prevailing standards to withstand the maximum amperage of the outdoor unit
   - Take into account the distance between the appliances and the electrical panel
   - Take into account the earthing system
6. Supply the appliance via a circuit with an omnipolar switch with an opening gap distance of more than 3 mm. The installation must be equipped with a main switch.
7. Check the compliance of the protective earthing before making any electrical connections.

6.9.2 Precautions for electrical wiring

Important
After power off, wait for 5 minutes to power on again.

- It is prohibited to install emergency stop switches, remote switches for stopping the unit, including circuit breakers, contactors and relays, within 2 metres of the unit.
- The wiring must be carried out in accordance with the wiring diagram supplied with the unit and the instructions given below.
- Use a dedicated power source. Do not connect the unit to a power supply cable, surge protector, or telephone earth.
- Connect the outdoor unit power supply directly to the electrical panel without an additional filter.
  It is prohibited to install a phase rectifier.
- Install a residual-current device (30 mA) to prevent any electric shock.
- Install a residual-current device (RCD) compatible with high harmonics for the power supply to the "inverter" outdoor unit to avoid unnecessary tripping.
- Install the necessary circuit breakers.

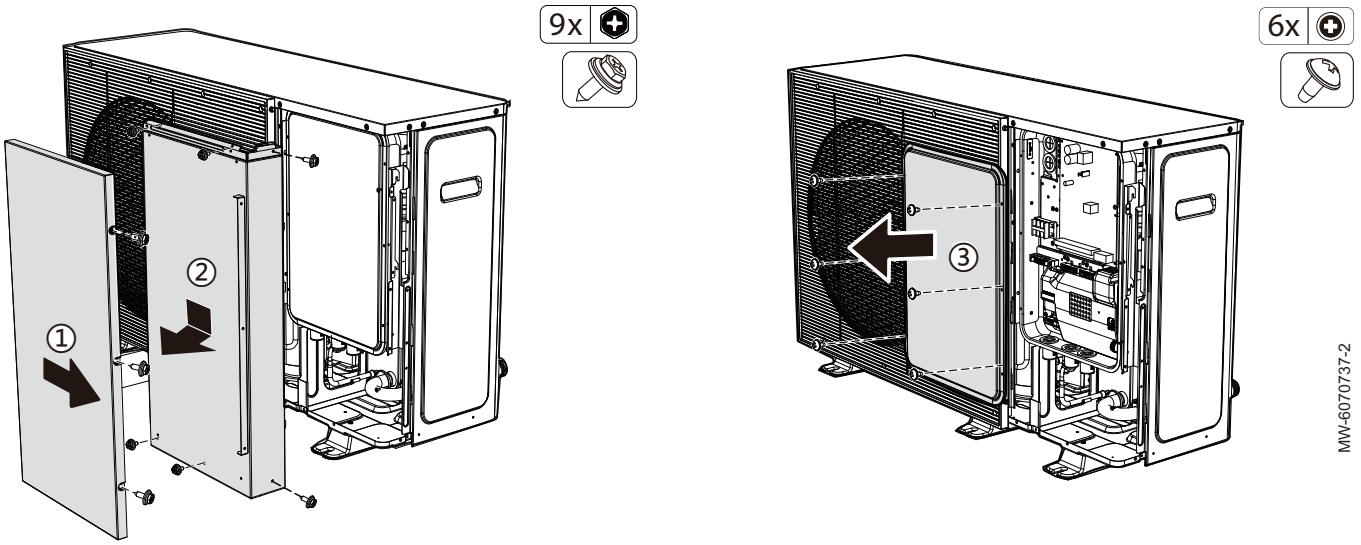
Power supply cable and communication cable

- Use only copper wires.
- Use shielded communication cables, including the XYE line between the outdoor unit and the indoor unit.
- Use a H07RN-F double-insulated multi-strand cable for the power supply.
- Route power supply cables and communication cables separately. Power supply cables and communication cables should not be placed in the same sheath to avoid electromagnetic interference.
- Never compress wire harnesses and ensure that wiring is kept away from pipes and sharp edges.
- Secure electrical cables with cable clamps.
- Make sure no external pressure is applied to the terminal connections.

6.9.3 Accessing the main control box

To access the PCBs and electrical terminal blocks, follow the instructions below.

Fig.44



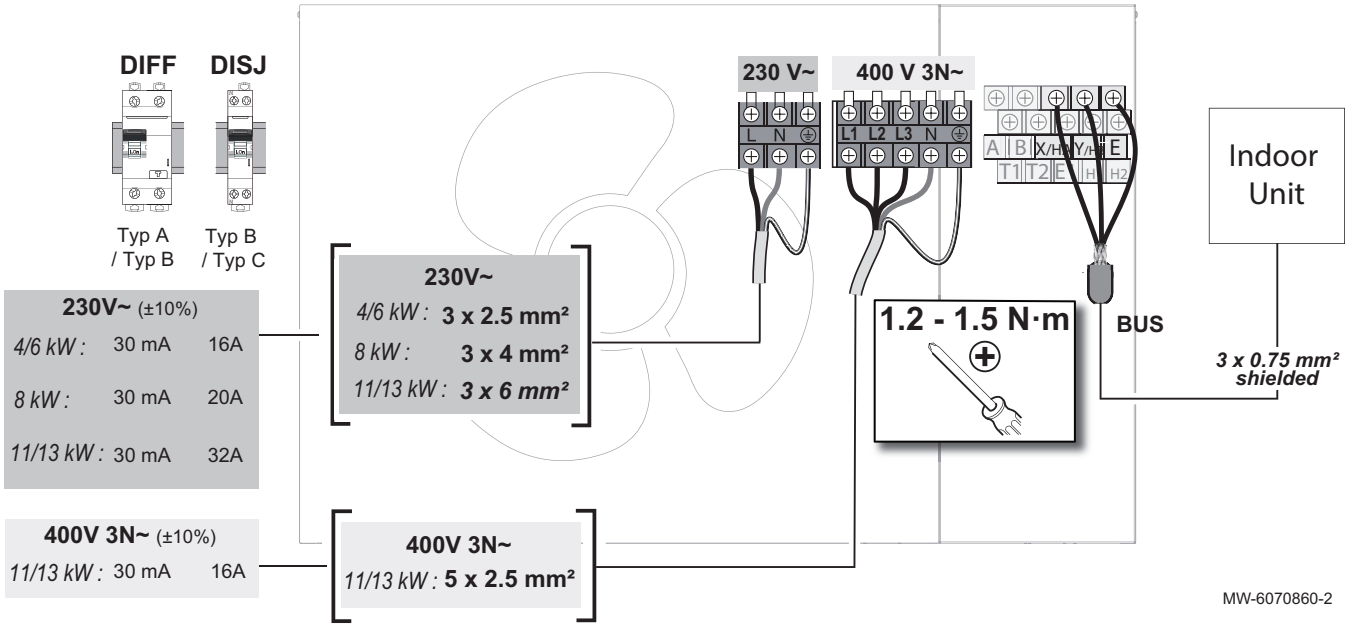
Warning
- Risk of electric shock.
- Risk of scalding.

1. Remove the front panel.
2. Remove the inner panel.

3. Remove the main control box cover.

6.9.4 Connecting the electrical circuits

Fig.45



The cable cross-sections are given as a guide only.

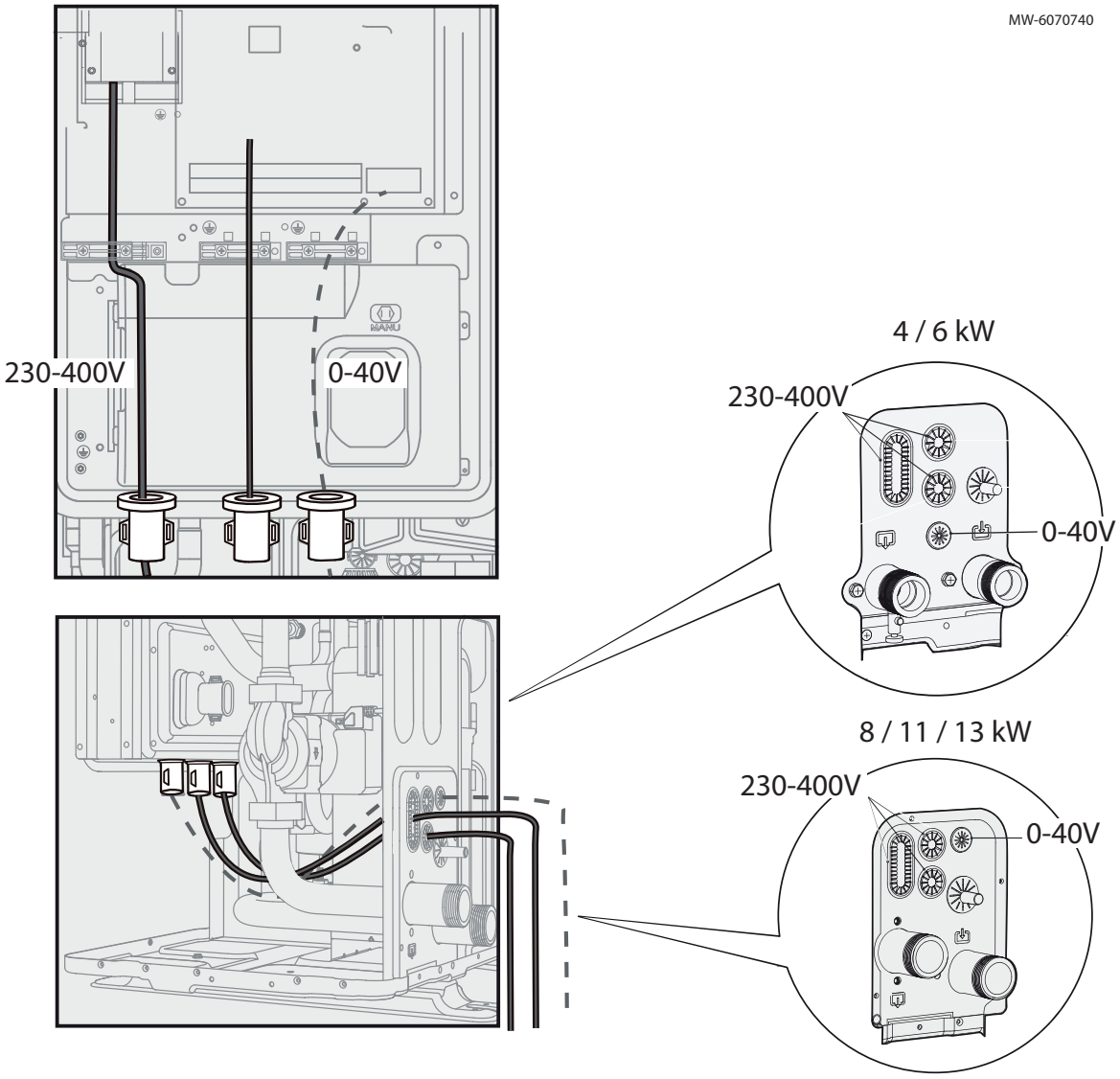
Important
Use a shielded cable for the BUS connection between the indoor unit and the outdoor unit to avoid communication problems.

BUS	Outdoor unit connection bus
DIFF	Residual-current device (RCD)
DISJ	Protection circuit breaker
shielded	Shielded cable for connection with the outdoor unit
Indoor Unit	Indoor unit

6.9.5 Cable routing

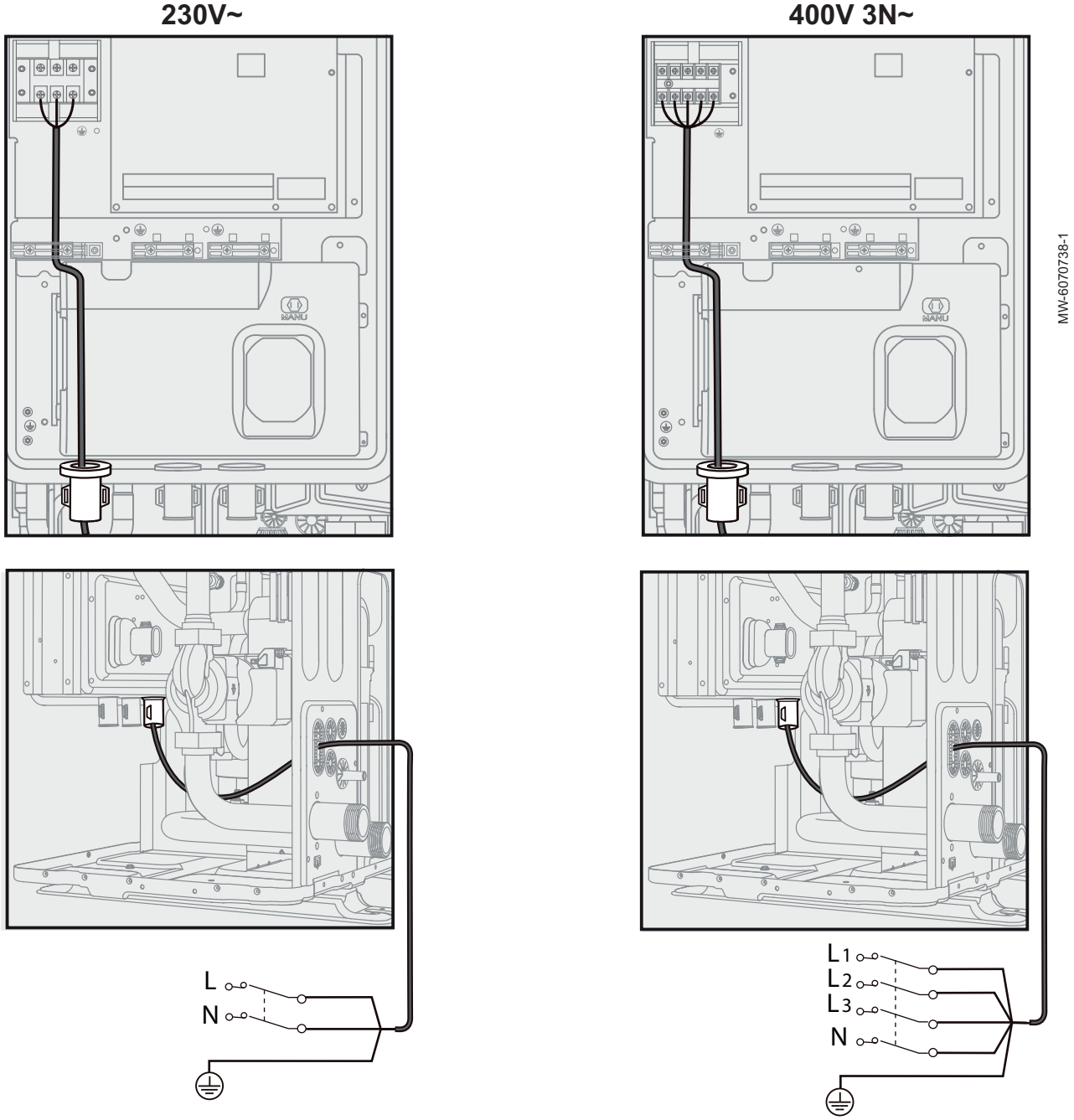
The 0-40 V cables must be separated from the 230/400 V cables.

Fig.46


MW-6070740
230-400V
0-40V
4 / 6 kW
230-400V
0-40V
8 / 11 / 13 kW
230-400V
0-40V

6.9.6 Connecting the outdoor unit to the power supply

Fig.47



L1 L2 L3 L Live
N Neutral

⏚ Ground

It is possible to use a 230 V phase on a three-phase electric panel, in compliance with the applicable standards.

The electrical connection of the outdoor unit must be made via a dedicated circuit. Before connecting, check that the cross-section of the cable and the circuit breaker on the electric panel are suitable.

1. Connect the wires to the appropriate terminals.
2. To avoid electric shock, make sure that the length of the conductors between the cable clamp and the terminal blocks is such that the active conductors are put under tension before the earth conductor.

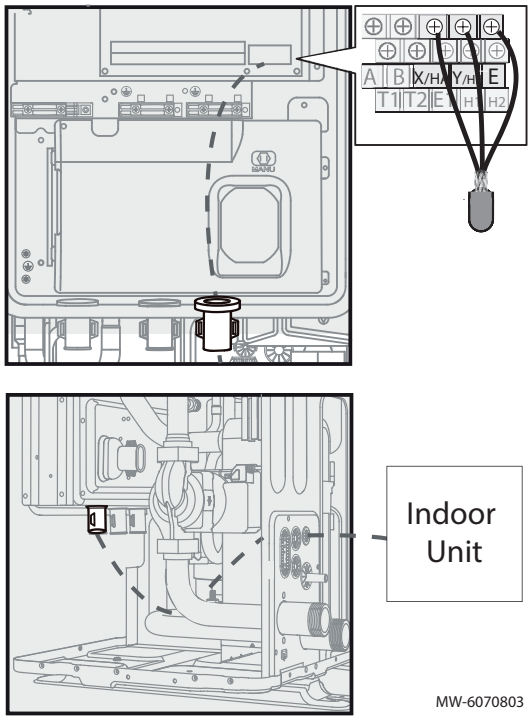
Danger
The earth wire must be 10 mm longer than the **N** and **L** wires.

3. Feed the cable into the traction arrester device.

4. Adapt the cable length.
5. Tighten the screws correctly.

6.9.7 Connect the outdoor unit to the indoor unit

Fig.48



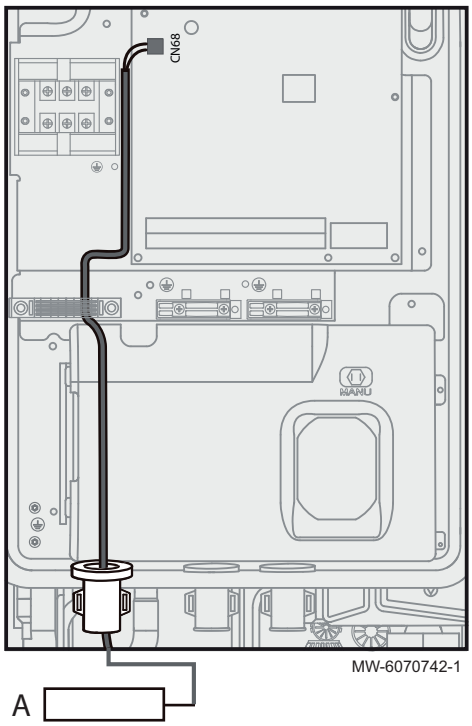
Indoor Unit Indoor unit

1. Connect the wires to the appropriate terminals.
2. Feed the cable into the cable clamp.
3. Adapt the cable length.
4. Tighten the screws correctly.

6.9.8 Connection for Other Optional Components

■ **Wiring of drainage pipe heating tape**

Fig.49



A Heating tape

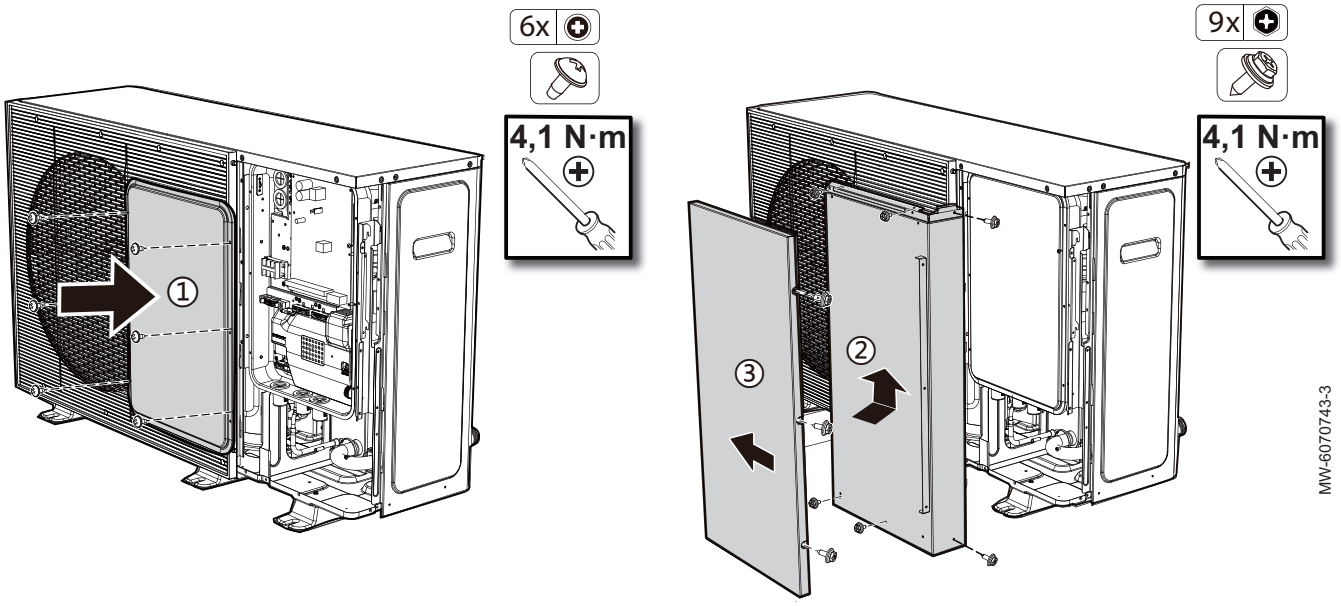
For the drainage pipe heating tape, the power shall not exceed 40 W. Working current is up to 200 mA, supply voltage 230 VAC.

Use tie wraps

Important
To prevent refrigerant or insects from entering into the electrical control box and causing fire,after wiring, the sleeve should be fastened with a tie wrap (accessory)

6.10 Refit the outdoor unit panels

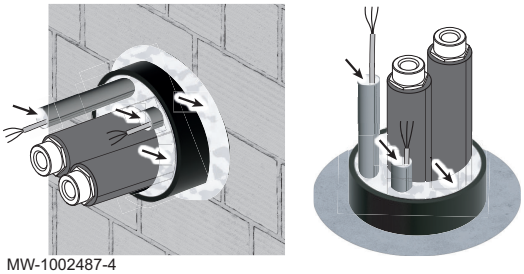
Fig.50



1. Install the main control box cover.
2. Install the inner panel.
3. Install the front panel.

6.11 Sealing building entry points

Fig.51


MW-1002487-4

The outdoor unit contains a highly flammable refrigerant. In the event of a leak in the refrigerant circuit, this refrigerant must not penetrate inside buildings.

Once the water and electrical connections have been made:

1. Check that the sensor cables (0-40 V) are separate from the 230/400 V cables.
2. Seal the wall openings.

Important
Do not place the connection tubes in direct contact with the wall to ensure no vibrations from the heat pump are transmitted to the wall.

3. Seal the ends of the ducts in which the electrical cables and heating pipes run.

6.12 Flushing and filling the heating installation with mains water

6.12.1 Flushing new installations and installations less than 6 months old

Before filling the heating installation, it is essential to remove any debris (copper, caulking, soldering flux) from the installation.

1. Clean the installation using a suitable cleaning agent.
2. Flush the system with at least 3 times the volume of water contained in the entire system (until the water runs clear and shows no impurities).
3. Check and clean the filters if necessary.

6.12.2 Flushing an existing installation

Before filling the heating installation, it is essential to remove any sludge deposits which have accumulated in the heating circuit over the years.

1. Remove any sludge from the installation.

2. Flush the installation with at least 3 times the volume of water contained in the central heating system (until the water runs clear and shows no impurities).
3. Check and clean the filters if necessary.

6.12.3 Filling the heating circuit with mains water

After flushing the installation and checking that the filters are clean, the heating circuit must be filled with mains water.

1. Fill the system until it reaches a pressure of between 0.15 and 0.2 MPa (1.5 and 2 bar).
   Read the pressure using a pressure gauge.
2. Check the leak-tightness of the heating circuit.
3. Vent the installation fully for optimal operation.



See also
Heating water specifications, page 41

6.13 Flushing and filling a heating installation with glycol water

6.13.1 Flushing an installation before filling with glycol water

Before filling the heating installation, you must remove any debris from the installation (copper, caulking, soldering flux) and any sludge deposits which have accumulated in the heating circuit (for existing installations).

1. Clean the circuit in accordance with the glycol manufacturer's directives.
2. Check and clean the filters if necessary.

6.13.2 Filling the heating circuit with glycol water



Important
Refer to the local regulations and the system components manuals to check the compatibility.

Glycol lowers the freezing point of water.



Caution
Ethylene glycol and propylene glycol are toxic.



Caution
Glycol can corrode the system. When uninhibited glycol comes into contact with oxygen, it becomes acidic. This corrosion process is accelerated by copper and high temperature. The acidic uninhibited glycol attacks metal surfaces, forming galvanic corrosion cells that can cause severe damage to the system. Therefore, it is important to follow these steps:

- Let a qualified specialist treat the water correctly;
- Select a glycol with corrosion inhibitors to counteract acids formed by the oxidation of glycols;
- Do not use any automotive glycol because its corrosion inhibitors have a limited lifetime and contain silicates which can contaminate or block the system;
- Do not use galvanized pipes in glycol systems as these may lead to the precipitation of certain components in the glycol's corrosion inhibitor.



Important
Glycol absorbs moisture from the environment, so it is important to avoid using glycol exposed to air. If glycol is left uncovered, the water content increases, lowering the glycol concentration and potentially causing hydraulic components to freeze. To prevent this, take precautions and minimize glycol’s exposure to air.

The types of glycol that can be used depend on whether the system contains a domestic hot water tank:

Tab.29

If	Then
The system contains a domestic hot water cylinder	Only use propylene glycol(1)
The system does NOT contain a domestic hot water cylinder	Either propylene glycol(1) or ethylene glycol can be used
(1) Propylene glycol, including the necessary inhibitors, falls in Category III according to EN 1717.	

The required concentration of glycol depends on the lowest expected outdoor temperature, and on whether you want to protect the system from bursting or from freezing. To prevent the system from freezing, more glycol is required.

Add glycol according to the table below.

Lowest expected outdoor temperature	Prevention from bursting(1)	Prevention from freezing(2)
–5 °C	10%	15%
–10 °C	15%	25%
–15 °C	20%	35%
–20 °C	25%	N/A(3)
–25 °C	30%	N/A(3)
–30 °C	35%	N/A(3)
(1) Glycol can prevent the piping from bursting, but cannot prevent the liquid inside the piping from freezing. (2) Glycol can prevent the liquid inside the piping from freezing. (3) Additional action is needed to prevent freezing.		

Important

- The required concentration might vary depending on the type of glycol used. ALWAYS compare the requirements from the table above with the specifications provided by the glycol manufacturer. If necessary, meet the requirements set by the glycol manufacturer.
- The added concentration of glycol should NEVER exceed 35%.
- If the liquid in the system is frozen, the pump will NOT be able to start. Please note that solely preventing the system from bursting may not prevent the liquid inside from freezing.
- If water remains stagnant within the system, it is highly likely to freeze and result in system damage.

Adding glycol to the water circuit reduces the maximum allowed water volume of the system

6.13.3 Filling the heating circuit with glycol water

1. Fill the installation using a glycol water mix filling station until a pressure of 0.15 to 0.2 MPa (1.5 to 2 bar) is achieved.
2. Check the leak-tightness of all the fittings and gaskets on the heating circuit.
3. Fully vent the outdoor unit and the installation before commissioning.
4. Fully vent the outdoor unit and the installation once more after several hours running at a high operating temperature.
   ⇨ This removes any small air bubbles formed in the glycol water at high operating temperatures.
5. Once more visually check the leak-tightness of all fittings and gaskets on the heating circuit after several hours running at a high operating pressure.
   Monopropylene glycol leaks more easily than water. The pressure tests do not guarantee that there will be no leaks when the installation is filled with glycol water under pressure.

See also
Heating water specifications, page 41

6.14 Checking water circuit

The following conditions must be met:

- The maximum water pressure is lower than or equal to 3 bar.
- The maximum water temperature is lower than or equal to 75 °C depending on the safety device setting.
- Drain valves must be installed at the lowest points of the system to ensure complete drainage of the circuit during maintenance.
- Bleed valves must be installed at the highest points of the system. The bleed valves must be located at points that are easily accessible for service.

7 User instructions

7.1 Precautions regarding the location of the outdoor unit

The outdoor unit contains a highly flammable refrigerant.

Observe the precautions relating to the location of the outdoor unit throughout the service life of the product:

- Maintain free air circulation around the outdoor unit (intake and outlet).
- Do not install anything in the protective perimeter around the outdoor unit (for example, a window, garden shed, parking spot or outdoor lighting).
- Keep any potential source of sparks (for example, lit cigarette, mower, barbecue or parked car) away from the protective perimeter.
- Do not store or handle corrosive products in the vicinity of the outdoor unit (for example, aerosols, chlorinated cleaning products, paints or road salt).

7.2 Prolonged power outage in winter

A prolonged power outage at outdoor temperatures below zero may damage your heating installation. Action may be required, depending on the configuration of your installation:

- Installation with glycol: no action required.
- Installation equipped with anti-freeze valves: drain the outdoor part of the heating installation during a prolonged power outage (10 hours or more) at outdoor temperatures below zero.
- Installation without glycol and not equipped with anti-freeze valves: drain the outdoor part of the heating installation.

Important
If in doubt about the draining and filling procedure, contact your installer.

7.3 Clearing the snow accumulated around the outdoor unit

Snow accumulated around the outdoor unit may impair its performance.

1. Regularly check that there is no snow accumulated around the outdoor unit.
2. Clear snow, if necessary.

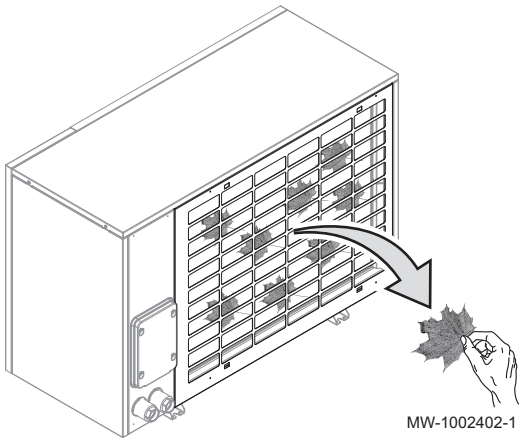
7.4 Cleaning the outdoor unit

Clogging by dust and other particles impairs the heat pump's performance.

Check the cleanliness of the heat pump twice a year.

Caution
Do not clean the heat pump with a high-pressure cleaner.

Fig.52



1. Remove any leaves caught in the fins at the back of the appliance.

Danger
Risk of injury on the sharp edges of the fins at the back on the appliance.

Caution
Do not distort or damage the fins.

2. Remove the dust and debris with a soft-bristled brush or hand broom.
3. Clean the outside of the outdoor unit using a damp cloth and a mild detergent.

8 Maintenance

Regular checks and inspections at certain intervals are required to guarantee the optimal performance of the unit.

8.1 Safety precautions for maintenance

Danger
Risk of electric shock.

Warning
- Please note that some parts of the electric component box are hot.
- To avoid the risk of electric shock or fire, do not spray the outdoor unit with water.
- Do not leave the outdoor unit unattended when the casing is removed.

Important
- Before performing any maintenance or service work, touch a metal part of the unit to eliminate static electricity and to protect the PCB.
- Without regular maintenance, the performance of the unit may degrade and the risk of parts damage may increase gradually.

8.2 Maintenance checklist

Tab.30 By installer

Items	Recommended frequency
General	
Check that all the parts are in the proper position.	Once a year
Clean the evaporator	Once a year
Hydraulic circuit	
Check that the water pressure is sufficient.	Once a year
Clean the strainer in the water system.	Once a year
Check that the flow switch is working properly.	Once a year
Check that the safety pressure relief valve (in the water system) is working properly.	Once a year
Check that the safety pressure relief valve (in the DHW water loop) is working properly.	Once a year

Items	Recommended frequency
Check that there is no water leakage in the water loop.	Once a year
Check that the frost protection solution is present and in good condition.	Once a year
Check that the backup heater of the domestic hot water cylinder is clean and in good condition.	Once a year
Wiring and electrical parts	
Check that the temperature sensor is working properly.	Once a year
Check that the wiring and cables of the installation are in good condition.	Once a year
Check that the contactors and circuit breakers are working properly.	Once a year
Refrigerant loop	
Check that there is no refrigerant leakage in the refrigerant loop.	Once a year

8.3 Service information

8.3.1 Label for refrigerant presence

Ensure that proper labels are pasted on the equipment stating the equipment contains flammable refrigerant.

8.3.2 Leak detection methods

The following leak detection methods are deemed acceptable for systems containing flammable refrigerants. An electronic leak detector should be used to detect flammable refrigerants, but its sensitivity may not be adequate, or the detector may need recalibration. (Detection equipment should be calibrated in a refrigerant-free area.) Ensure that the detector is not a potential source of ignition and is suitable for the refrigerant. Leak detection equipment should be set at a percentage of the LFL of the refrigerant and should be calibrated to be suitable for the refrigerant employed. The appropriate percentage of gas (25 % maximum) is confirmed. Leak detection fluids are suitable for use with most refrigerants but detergents containing chlorine should not be used as the chlorine may react with the refrigerant and corrode the copper pipes. If a leak is suspected, all naked flames should be removed or extinguished. If a leakage of refrigerant is found and brazing is required, all of the refrigerant should be recovered from the system, or isolated (by means of shut off valves) in a part of the system that is remote from the leak. Oxygen free nitrogen (OFN) should then be purged through the system both before and during the brazing process.

8.3.3 Check of refrigeration equipment

Where electrical components are to be changed, they should be fit for the intended purpose and comply with the correct specifications. Always follow the manufacturer’s maintenance and service guidelines. In case of any doubt, consult the manufacturer’s technical department for assistance. Check installations using flammable refrigerants.

- The amount of refrigerant to be charged depends on the size of the room where the refrigerant-containing parts are installed.
- The ventilation machinery and outlets should work adequately and be not obstructed.
- If an indirect refrigerating circuit is used, the secondary circuits should be checked for any refrigerant; Markings on the equipment should be visible and legible.
- Illegible markings and signs should be corrected.
- Refrigeration pipes or components should be installed in apositions where they are unlikely to be exposed to any substance that may corrode refrigerant-containing components, unless the components are constructed of materials that are inherently resistant to corrosion or are suitably protected from corrosion.

8.3.4 Check of electrical devices

Repair and maintenance of electrical components should include initial safety checks and component inspection procedures. If a fault exists and could compromise safety, no electrical supply should be connected to the circuit until it is satisfactorily dealt with. If the fault cannot be corrected immediately but it is necessary to continue operation, an adequate temporary solution should be adopted. This should be reported to the owner of the equipment so all parties are advised.

Initial safety checks should include the following:

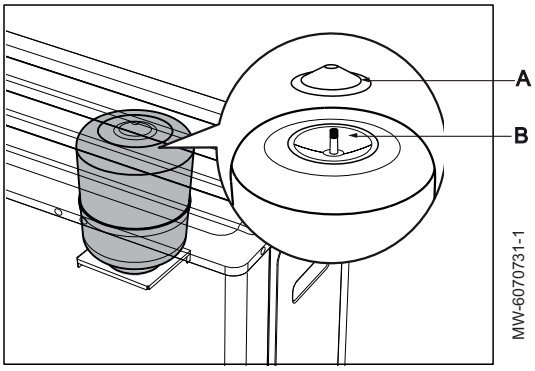
- The capacitors should be discharged in a safe manner to avoid sparking risks.
- No live electrical components and wiring can be exposed during the system charging, recovery or purging.
- Earth bonding should be continuous.

8.3.5 Pre-pressure adjustment of the expansion vessel

The outdoor unit is equipped with an expansion vessel (5 effective litres) that has a default pre-pressure of 0.15 MPa (1.5 bar).

Check the pre-pressure every two years.

Fig.53



A Top cover
B Schrader valve

9 Error codes

Tab.31 Water circuit errors

Error code	Description	Displayed on
E0	Water flow error (10 times E8)	User interface and Main Control PCB
E8	Water flow protection	User interface and Main Control PCB

Tab.32 Communication errors

Error code	Description	Displayed on
CL	Water pump communication wire fault	User interface and Main Control PCB
E2	Communication fault between user interface and Main Control PCB	User interface and Main Control PCB
EL	Communication fault between hydraulic module and MH kit	User interface and Main Control PCB
H0	Communication error on Main Control PCB	User interface and Main Control PCB
H1	Communication error between Main Control PCB and inverter PCB	User interface and Main Control PCB
Hd	Communication fault between the master outdoor unit and the slave outdoor unit.	User interface and Main Control PCB

Tab.33 Sensor errors

Error code	Description	Displayed on
E3	T1 Electric Heater/AHS water outlet temperature sensor error	User interface and Main Control PCB
E4	T5 water cylinder temperature sensor error	User interface and Main Control PCB
E5	T3 outdoor unit heat exchanger bottom temperature sensor error	User interface and Main Control PCB
E6	T4 room temperature sensor error	User interface and Main Control PCB
E7	Tbt buffer tank temperature sensor/cascade system final outlet water temperature sensor error	User interface and Main Control PCB
E9	Th return air temperature sensor error	User interface and Main Control PCB
EA	Tp discharge temperature sensor error	User interface and Main Control PCB
Eb	Tsolar solar panel temperature sensor error	User interface and Main Control PCB
EC	T5_2 water cylinder temperature sensor error	User interface and Main Control PCB
Ed	Tw_in plate heat exchanger inlet water temperature sensor error	User interface and Main Control PCB
FC1	TL outdoor unit heat exchanger outlet temperature sensor error	User interface and Main Control PCB
H2	T2 plate heat exchanger outlet refrigerant temperature sensor error	User interface and Main Control PCB
H3	T2B plate heat exchanger inlet refrigerant temperature sensor error	User interface and Main Control PCB
H5	Ta room temperature sensor error	User interface and Main Control PCB
H8	H-SEN high pressure sensor error	User interface and Main Control PCB
H9	Tw2 zone 2 water flow temperature sensor error	User interface and Main Control PCB
HA	Tw_out plate heat exchanger outlet water temperature sensor error	User interface and Main Control PCB
PF	Water flow sensor failure	User interface and Main Control PCB
PE	Water pressure sensor failure	User interface and Main Control PCB
P21	L-SEN low pressure sensor error	User interface and Main Control PCB
P27	H-SEN and L-SEN connected in reverse (detect when the compressor is off)	User interface and Main Control PCB

Tab.34 Voltage errors

Error code	Description	Displayed on
E1	Phase loss or phase reversal	User interface and Main Control PCB
H7	Power overvoltage and Power undervoltage protection	User interface and Main Control PCB

Tab.35 Inverter module errors

Error code	Description	Displayed on
F1	Protection against DC bus undervoltage	User interface and Main Control PCB
C7	Excessive temperature protection for IPM module	User interface and Main Control PCB
H4	3 times “L1*” in 60 mins	User interface and Main Control PCB
L1E	Hardware overcurrent protection	Main Control PCB
L11	Phase current instantaneous overcurrent protection	Main Control PCB
L12	Phase current continuous 30 s overcurrent protection	Main Control PCB
L2E	Overheating protection	Main Control PCB
L3E	Bus voltage too low error	Main Control PCB
L31	Bus voltage too high error	Main Control PCB
L32	Bus voltage excessively high error	Main Control PCB
L34	Phase loss error on the three-phase power supply	Main Control PCB
L43	Abnormal phase current sampling bias	Main Control PCB

Error code	Description	Displayed on
L45	Fan motor code mismatch error	Main Control PCB
L46	IPM protection (FO)	Main Control PCB
L47	Module type mismatch	Main Control PCB
L5E	Motor failed to start	Main Control PCB
L52	Motor stalling protection	Main Control PCB
L6E	Phase loss protection	Main Control PCB
L61	Compressor terminal short circuit protection	Main Control PCB
L65	IPM short circuit protection	Main Control PCB
LBE	Action of high pressure switch	Main Control PCB
LB7	PED error	Main Control PCB
LCE	PFC hardware overcurrent protection	Main Control PCB
LC1	Instantaneous overcurrent of PFC software protection	Main Control PCB
LC2	PFC software continuous 30 s overcurrent protection	Main Control PCB
LC3	PFC low voltage protection	Main Control PCB
LC4	PFC power factor is less than 0.8	Main Control PCB
LC5	PFC valid value overcurrent protection	Main Control PCB
LC6	PFC1 channel hardware overcurrent protection	Main Control PCB
LC7	PFC2 channel hardware overcurrent protection	Main Control PCB
LC8	PFC3 channel hardware overcurrent protection	Main Control PCB
LC9	Overheating protection of PFC module	Main Control PCB
LCA	PFC module CBC overcurrent error protection	Main Control PCB
LCB	Overvoltage of PFC bus or PFC half bus	Main Control PCB
LCC	Short circuit of PFC IGBT	Main Control PCB
LCD	Abnormal PFC sampling bias	Main Control PCB
H6	Fan failure	User interface and Main Control PCB
HH	10 times H6 in 120 mins	User interface and Main Control PCB
J1E	Hardware overcurrent protection	Main Control PCB
J11	Phase current instantaneous overcurrent protection	Main Control PCB
J12	Phase current continuous 30 s overcurrent protection	Main Control PCB
J2E	Overheating protection	Main Control PCB
J3E	Bus voltage too low error	Main Control PCB
J31	Bus voltage too high error	Main Control PCB
J32	Bus voltage excessively high error	Main Control PCB
J43	Abnormal phase current sampling bias	Main Control PCB
J45	Fan motor code mismatch error	Main Control PCB
J46	IPM protection (FO)	Main Control PCB
J5E	Motor failed to start	Main Control PCB
J52	Motor stalling protection	Main Control PCB
J6E	Phase loss protection	Main Control PCB
J61	Fan terminal short circuit protection	Main Control PCB
J65	IPM short circuit protection	Main Control PCB
HF	Outdoor unit EEPROM error	User interface and Main Control PCB

Tab.36 Others

Error code	Description	Displayed on
F6	EXV fault	Main Control PCB
F75	Insufficient discharge superheating	Main Control PCB
Pd	Frost protection operation	Main Control PCB

Tab.37 Protection code

Error code	Description	Displayed on
P0	Low pressure protection	User interface and Main Control PCB
P1	High pressure switch protection	User interface and Main Control PCB
P3	Overcurrent protection	User interface and Main Control PCB
P4	Protection against excessive compressor discharge temperature	User interface and Main Control PCB
Pd	Protection against excessive condensation temperature	User interface and Main Control PCB
HP	Low pressure protection in cooling mode	User interface and Main Control PCB
bA	T4 sensor out of operation range protection	User interface and Main Control PCB
PP	Protection against an abnormal temperature difference between the water outlet and inlet	User interface and Main Control PCB
Hb	3 times PP and Twout < 7 ℃	User interface and Main Control PCB
P5	Abnormal temperature difference between the water outlet and inlet temperature	User interface and Main Control PCB

10 Decommissioning and disposal

10.1 Disposal and recycling

Fig.54



MW-3000179-03

Warning
Removal and disposal of the outdoor unit must be carried out by a qualified professional in accordance with prevailing local and national regulations.

1. Switch off the indoor and outdoor units.
2. Shut off the electrical power supply to the outdoor and indoor units.
3. Recover the refrigerant in accordance with prevailing regulations
4. Close the water mains.
5. Drain the installation.
6. Dismantle all hydraulic connections.
7. Remove the outdoor unit.
8. Scrap or recycle the outdoor unit in accordance with prevailing local and national regulations.

10.2 Brine disposal/recycling

Particular care must be taken in the collection and processing of hazardous waste. It must be disposed of in accordance with the regulatory requirements governing special waste.

Do not dispose of water containing monopropylene glycol into the sewage system or the natural environment. Hazardous substances may penetrate the water tables or enter the food chain, causing harmful effects for health and well-being. Place a container of sufficient volume under the drain pipe to collect waste. Store waste and contaminated products in specially designed closed and sealed containers. Dispose of them in accordance with applicable regulations or send them to a waste recycling centre for processing.

Important
Always comply with applicable local regulations concerning correct recycling of special waste.

10.3 Recover refrigerants

Caution
It is recommended to wear protective gloves and goggles before carrying out any work on the refrigerant circuit.

When decommissioning the heat pump, all refrigerants need to be recovered safely. Prior to carrying out the recovery operation, an oil and refrigerant sample must be taken in case analysis is required prior to re-use of recovered refrigerant. It is essential that electrical power is switched off before the task is commenced.

Before carrying out the procedure, ensure that:

- Mechanical handling equipment is available, if required, for handling refrigerant cylinders
- All personal protective equipment is available and being used correctly
- The recovery process is supervised at all times by a competent person
- Recovery equipment and cylinders conform to the appropriate standards

1. Become familiar with the equipment and its operation.
2. Isolate the system electrically.
3. Pump down the refrigerant system, if possible.
4. If a vacuum is not possible, fit a manifold so that refrigerant can be removed from various parts of the system.
5. Make sure that cylinder is placed on the scales before starting recovery.
6. Start the recovery machine and run it following the instructions.

Important
- Do not overfill the cylinders (the liquid charge must not exceed 80% volume).
- Do not exceed the maximum working pressure of the cylinder, even temporarily.

7. When the cylinders have been filled correctly and the process completed, make sure that the cylinders and the equipment are removed from site promptly and all isolation valves on the equipment are closed off.

Important
Recovered refrigerant shall not be charged into another refrigerating system unless it has been cleaned and checked.

10.4 Labelling

Equipment shall be labelled stating that it has been de-commissioned and emptied of refrigerant. The label shall be dated and signed.

10.5 Recovery equipment

When removing refrigerant from a system, either for servicing or decommissioning, it is recommended good practice that all refrigerants are removed safely.

When transferring refrigerant into cylinders, ensure that only appropriate refrigerant recovery cylinders are employed. Ensure that the correct number of cylinders for holding the total system charge is available. All cylinders to be used are designated for the recovered refrigerant and labelled for that refrigerant (i.e. special cylinders for the recovery of refrigerant). Cylinders shall be complete with pressure-relief valve and associated shut-off valves in good working order. Empty recovery cylinders are evacuated and, if possible, cooled before recovery occurs.

The recovery equipment shall be in good working order with a set of instructions concerning the equipment that is at hand and shall be suitable for the recovery of all appropriate refrigerants including, when applicable, flammable refrigerants. In addition, a set of calibrated weighing scales

shall be available and in good working order. Hoses shall be complete with leak-free disconnect couplings and in good condition. Before using the recovery machine, check that it is in satisfactory working order, has been properly maintained and that any associated electrical components are sealed to prevent ignition in the event of a refrigerant release. Consult manufacturer if in doubt.

The recovered refrigerant shall be returned to the refrigerant supplier in the correct recovery cylinder, and the relevant waste transfer note arranged. Do not mix refrigerants in recovery units and especially not in cylinders.

If compressors or compressor oils are to be removed, ensure that they have been evacuated to an acceptable level to make certain that flammable refrigerant does not remain within the lubricant. The evacuation process shall be carried out prior to returning the compressor to the suppliers. Only electric heating to the compressor body shall be employed to accelerate this process. When oil is drained from a system, it shall be carried out safely.

11 Appendix

11.1 Benchmark Commissioning & Warranty Validation Service Record

Fig.55



AD-3000725-01

Fig.56



MW-6070892

It is a requirement that the heat pump is installed and commissioned to the manufacturers' instructions and the data fields on the commissioning checklist completed in full.

To instigate the warranty the heat pump needs to be registered with the manufacturer within one month of the installation. The warranty rests with the end-user (consumer), and they should be made aware it is ultimately their responsibility to register with the manufacturer, within the allotted time period.

It is essential that the heat pump is serviced in line with the manufacturers' recommendations, at least annually. This must be carried out by a competent, certified operative. The service details should be recorded on the Benchmark Service and Interim Heat Pump Work Record and left with the householder. Failure to comply with the manufacturers' servicing instructions and requirements will invalidate the warranty.

This Commissioning Checklist is to be completed in full by the competent person who commissioned the heat pump and associated equipment as a means of demonstrating compliance with the appropriate Building Regulations and then handed to the customer to keep for future reference.

Failure to install and commission according to the manufacturers' instructions and complete this Benchmark Commissioning Checklist will invalidate the warranty. This does not affect the customer's statutory rights.

All installations in England and Wales must be notified to Local Authority Building Control (LABC) either directly or through a Competent Persons Scheme. A Building Regulations Compliance Certificate will then be issued to the customer.

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11.2 Service record

It is recommended that your heating system is serviced regularly and that the appropriate Service Interval Record is completed.

Service provider
Before completing the appropriate Service Record below, please ensure you have carried out the service as described in the manufacturer's instructions. Always use the manufacturer's specified spare part when replacing controls.

SERVICE 01	Date:
Engineer name:	
Company name:	
Telephone No:	
Operative ID No:	
Comments:	
Signature:	

SERVICE 02	Date:
Engineer name:	
Company name:	
Telephone No:	
Operative ID No:	
Comments:	
Signature:	

SERVICE 03	Date:
Engineer name:	
Company name:	
Telephone No:	
Operative ID No:	
Comments:	
Signature:	

SERVICE 04	Date:
Engineer name:	
Company name:	
Telephone No:	
Operative ID No:	
Comments:	
Signature:	

SERVICE 05	Date:
Engineer name:	
Company name:	
Telephone No:	
Operative ID No:	
Comments:	
Signature:	

SERVICE 06	Date:
Engineer name:	
Company name:	
Telephone No:	
Operative ID No:	
Comments:	
Signature:	

* All installations in England and Wales must be notified to Local Authority Building Control (LABC) either directly or through a Competent Persons Scheme. A Building Regulations Compliance Certificate will then be issued to the custome .

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benchmark COLLECTIVE MARK
THE MARK OF QUALITY FOR THE INSTALLATION, COMMISSIONING AND SERVICING OF DOMESTIC HEATING AND HOT WATER SYSTEMS

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It is recommended that your heating system is serviced regularly and that the appropriate Service Interval Record is completed.

Service provider
Before completing the appropriate Service Record below, please ensure you have carried out the service as described in the manufacturer's instructions. Always use the manufacturer's specified spare part when replacing controls.

SERVICE 07	Date:
Engineer name:	
Company name:	
Telephone No:	
Operative ID No:	
Comments:	
Signature:	

SERVICE 08	Date:
Engineer name:	
Company name:	
Telephone No:	
Operative ID No:	
Comments:	
Signature:	

SERVICE 09	Date:
Engineer name:	
Company name:	
Telephone No:	
Operative ID No:	
Comments:	
Signature:	

SERVICE 10	Date:
Engineer name:	
Company name:	
Telephone No:	
Operative ID No:	
Comments:	
Signature:	

* All installations in England and Wales must be notified to Local Authority Building Control (LABC) either directly or through a Competent Persons Scheme. A Building Regulations Compliance Certificate will then be issued to the custome .

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11.3 Product data sheet

Tab.38 Product sheet for heat pump space heaters for medium-temperature applications: 55 °C

		Assure HP55-4-1-PHMB	Assure HP55-6-1-PHMB	Assure HP55-8-1-PHMB
Space heating energy efficiency class under average climate conditions		A+++	A+++	A+++
Rated heat output under average climate conditions *(Prated or Psup)*	kW	4.90	6.10	7.80
Seasonal space heating energy efficiency under average climate conditions	%	157	153	153
Annual energy consumption	kWh	2535	3233	4140
Sound power level L_{WA} indoors(1)	dB (A)	0	0	0
Rated heat output, under **colder - warmer** climate conditions	kW	4.30 - 4.70	5.90 - 6.00	8.00 - 8.80
Seasonal space heating energy efficiency, under **colder - warmer** climate conditions	%	124.3 - 170	132 - 179	136 - 188
Annual energy consumption **colder - warmer**	kWh	3328 - 1446	4325 - 1762	5659 - 2456
Sound power level L_{WA} outdoors	dB (A)	48	48	49
(1) If applicable				

Tab.39 Product sheet for heat pump space heaters for medium-temperature applications: 55 °C

		Assure HP55-11-1-PHMB	Assure HP55-11-3-PHMB	Assure HP55-13-1-PHMB	Assure HP55-13-3-PHMB
Space heating energy efficiency class under average climate conditions		A++	A++	A++	A++
Rated heat output under average climate conditions *(Prated or Psup)*	kW	12.10	12.10	13.70	13.70
Seasonal space heating energy efficiency under average climate conditions	%	147	147	146	146
Annual energy consumption	kWh	6662	6662	7588	7588
Sound power level L_{WA} indoors(1)	dB (A)	0	0	0	0
Rated heat output, under **colder - warmer** climate conditions	kW	10.80 - 12.40	10.80 - 12.40	13.90 - 14.90	13.90 - 14.90
Seasonal space heating energy efficiency, under **colder - warmer** climate conditions	%	127 - 174	127 - 174	128 - 181	128 - 181
Annual energy consumption **colder - warmer**	kWh	8197 - 3724	8197 - 3724	10408 - 4306	10408 - 4306
Sound power level L_{WA} outdoors	dB (A)	52	52	52	52
(1) If applicable					

See
For specific precautions about assembling, installing and maintaining: See Safety

11.4 Servicing information

To comply with regulations, the installer must fill these tables and hand them to the user. The user must keep this information safe for future reference.

Tab.40

	Emergency information
INSTALLER name and contact information	
INSTALLER SERVICING name and contact information	
FIRE DEPARTMENT contact information	
POLICE contact information	
LOCAL HOSPITAL contact information	
LOCAL BURNS UNIT contact information	

Tab.41

	Refrigerant information
Type of refrigerant	
Refrigerant formula	
Refrigerant flammability	
Maximum admissible pressure	

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